

Introduction to Docker

What is Docker?

- Docker is an open-source platform for developing, shipping, and running applications in containers.
- Ensures consistency across multiple environments.
- Lightweight and efficient compared to traditional virtual machines.

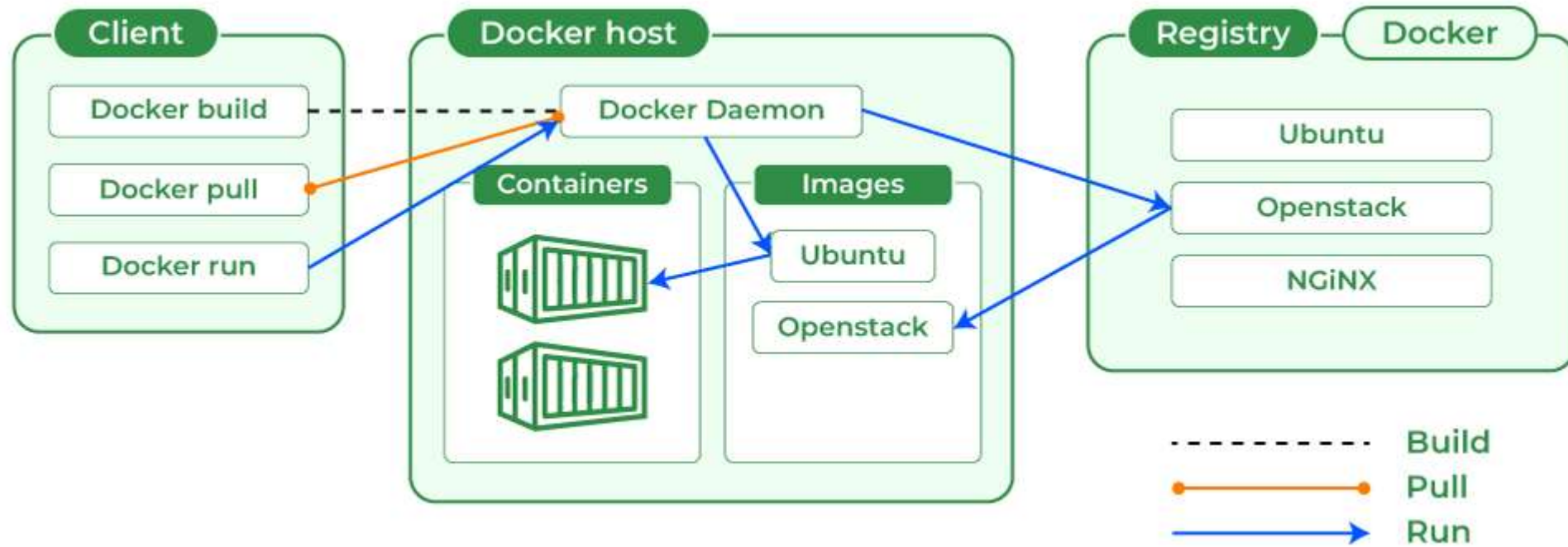
Why Use Docker?

- Fast and reliable deployment.
- Consistency across development, testing, and production environments.
- Efficient resource utilization.
- Easy scalability and portability.

Docker Architecture

- Docker Client: Command-line interface to interact with Docker.
- Docker Images: Read-only templates for creating containers.
- Docker Containers: Running instances of images.
- Docker Registry: Storage and distribution system for images (e.g., Docker Hub).

Docker Architecture



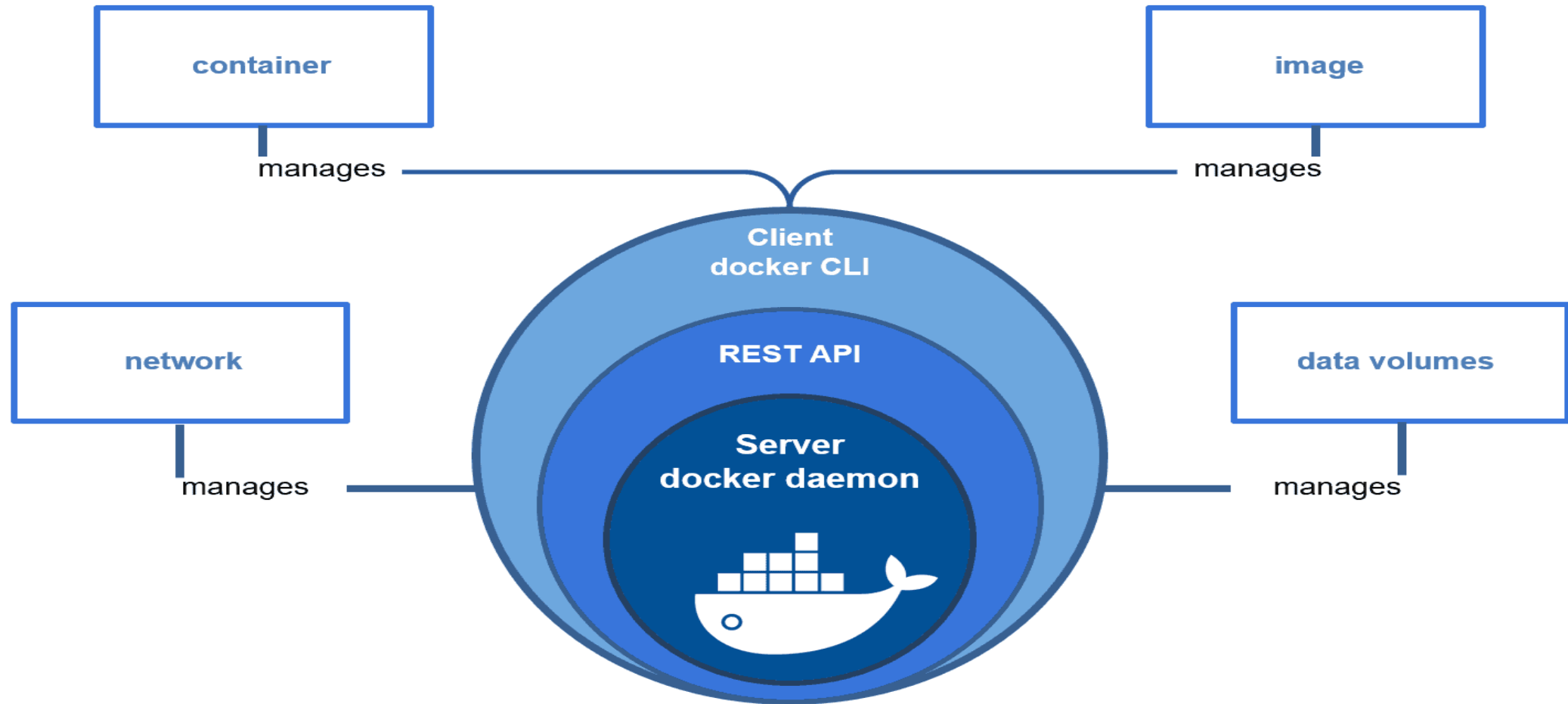
Docker vs Virtual Machines

Feature	Docker Containers	Virtual Machines
Startup Time	Seconds	Minutes
Resource Usage	Low	High
Portability	High	Limited
Isolation	Process-level	Full OS-level

Key Docker Components

- Dockerfile: Blueprint for building Docker images.
- Docker Compose: Tool for defining and running multi-container applications.
- Docker Swarm: Native clustering and orchestration tool.
- Kubernetes: Popular container orchestration system.

Key Docker Components



How to Install Docker

- Download and install Docker [from docker.com](https://docker.com).
- Install Docker Desktop for Windows/Mac.
- Use package manager for Linux (e.g., apt or yum).
- Verify installation with `docker --version`.



Cheatsheet for Docker CLI

Run a new Container

Start a new Container from an Image

```
docker run IMAGE  
docker run nginx
```

...and assign it a name

```
docker run --name CONTAINER IMAGE  
docker run --name web nginx
```

...and map a port

```
docker run -p HOSTPORT:CONTAINERPORT IMAGE  
docker run -p 8080:80 nginx
```

...and map all ports

```
docker run -P IMAGE  
docker run -P nginx
```

...and start container in background

```
docker run -d IMAGE  
docker run -d nginx
```

...and assign it a hostname

```
docker run --hostname HOSTNAME IMAGE  
docker run --hostname srv nginx
```

...and add a dns entry

```
docker run --add-host HOSTNAME:IP IMAGE
```

...and map a local directory into the container

```
docker run -v HOSTDIR:TARGETDIR IMAGE  
docker run -v ~/.:/usr/share/nginx/html nginx
```

...but change the entrypoint

```
docker run -it --entrypoint EXECUTABLE IMAGE  
docker run -it --entrypoint bash nginx
```

Manage Containers

Show a list of running containers

```
docker ps
```

Show a list of all containers

```
docker ps -a
```

Delete a container

```
docker rm CONTAINER  
docker rm web
```

Delete a running container

```
docker rm -f CONTAINER  
docker rm -f web
```

Delete stopped containers

```
docker container prune
```

Stop a running container

```
docker stop CONTAINER  
docker stop web
```

Start a stopped container

```
docker start CONTAINER  
docker start web
```

Copy a file from a container to the host

```
docker cp CONTAINER:SOURCE TARGET  
docker cp web:/index.html index.html
```

Copy a file from the host to a container

```
docker cp TARGET CONTAINER:SOURCE  
docker cp index.html web:/index.html
```

Start a shell inside a running container

```
docker exec -it CONTAINER EXECUTABLE  
docker exec -it web bash
```

Rename a container

```
docker rename OLD_NAME NEW_NAME  
docker rename 096 web
```

Create an image out of container

```
docker commit CONTAINER  
docker commit web
```

Manage Images

Download an image

```
docker pull IMAGE[:TAG]  
docker pull nginx
```

Upload an image to a repository

```
docker push IMAGE  
docker push myimage:1.0
```

Delete an image

```
docker rmi IMAGE
```

Show a list of all Images

```
docker images
```

Delete dangling images

```
docker image prune
```

Delete all unused images

```
docker image prune -a
```

Build an image from a Dockerfile

```
docker build DIRECTORY  
docker build .
```

Tag an image

```
docker tag IMAGE NEWIMAGE  
docker tag ubuntu ubuntu:18.04
```

Build and tag an image from a Dockerfile

```
docker build -t IMAGE DIRECTORY  
docker build -t myimage .
```

Save an image to .tar file

```
docker save IMAGE > FILE  
docker save nginx > nginx.tar
```

Load an image from a .tar file

```
docker load -i TARFILE  
docker load -i nginx.tar
```

Info & Stats

Show the logs of a container

```
docker logs CONTAINER  
docker logs web
```

Show stats of running containers

```
docker stats
```

Show processes of container

```
docker top CONTAINER  
docker top web
```

Show installed docker version

```
docker version
```

Get detailed info about an object

```
docker inspect NAME  
docker inspect nginx
```

Show all modified files in container

```
docker diff CONTAINER  
docker diff web
```

Show mapped ports of a container

```
docker port CONTAINER  
docker port web
```

Real World Use Cases

- Microservices architecture.
- CI/CD pipelines.
- Cloud-native applications.
- Development and testing environments.
- Hybrid and multi-cloud deployments.